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RESEARCH PAPER

The granular and polymer composite nature of kerogen-rich shale

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Abstract In the past decade, mechanical, physical, and chemical characterization of reservoir shale rocks, such as the Woodford shale, which is kerogen-rich shale (KRS), has moved toward micro- and nanoscale testing and analyses. Nanoindentation equipment is now widely used in many industrial and university laboratories to measure shale anisotropic Young's moduli, kerogen stiffness, plastic yield parameters, and other isotropic and anisotropic poromechanical and viscoelastic properties. However, to date, failure analyses of KRS and the effects of organic components on the tensile strength have not been observed or measured at the micro- or nanoscales. In this study, preserved kerogen-rich Woodford shale samples manufactured in micro-beam and micro-pillar geometries were mechanically tested and brought to failure in tension and compression, respectively. These tests were conducted in situ using a nanoindenter inside a scanning electron microscope (SEM). The load versus displacement curves of prismatic micro-cantilever beams were analyzed in light of high-resolution images collected during tensile fracture initiation, propagation, and ultimately sample failure. The micro-pillar geometries were subjected to a uniaxial compressive load and were also brought to failure while capturing measurements of stress and strain. It was found that, within just a few hundred microns of the KRS micro-cantilever beams, both brittle and ductile failure modes were observed. In the ductile plastic domain, strain-softening and strain-hardening behaviors were identified and characterized. These were not due to confining stress variations, but due to the volume of the organic matter and the way it is interlaced with the shale minerals in and around the failure planes. The tensile strength characteristics and the large modulus of toughness of kerogen, which is a cross-linked polymer, definitely weigh heavily in our engineering field applications, such as hydraulic fracking, which is a Mode I tensile fracture opening and propagation phenomenon. This practice demands that, due to the complex composite nature of KRS, mechanical characterization be not only for unconfined compressive strength but also for unconfined tensile strength and moduli of ruptures. At the end of this study, the need for nanometer scale mechanical characterization of KRS will become apparent. These nano- and micro-scale shale failure tests reinforce our previous understanding of the heterogeneous composite nature of Woodford KRS and its complex behavior, as well as other source shale reservoir formations.

Keywords Cantilever beams · Gas shale · Nano-/micro-failures · Pillars

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1 Introduction

Granular cohesionless earth materials with compaction histories, "memory" and compacted densities, are already very complex processes and still poorly understood when it comes to their mechanical property characterization [32,

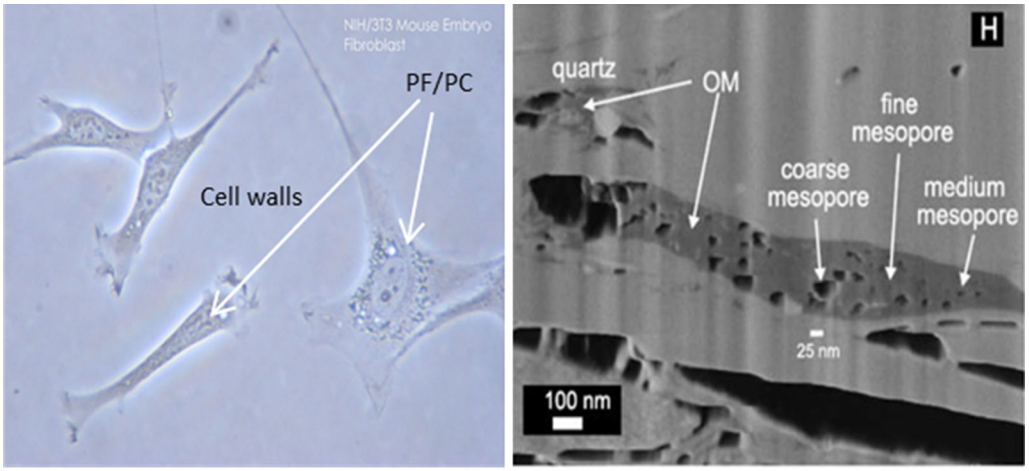
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50]. Clay-bearing sedimentary rocks, such as shale, formed under even more complex geological processes, are also mechanistically much less understood. The role of their mineral composition in the overall mechanical property characterization has been the subject of many studies [34, 60]. For more than a decade, now organic-rich shale has been the subject of micro- and nanomechanical characterization using the advances of nanotechnology. The early nanoindenters used on shale and mudstones were pioneered by Abousleiman and Ulm [3] and Zeszotarski et al. [61] while investigating the micro-mechanical behavior of shale rocks. The former study was interested in GeoGenoming™ clay and mudstones for applications in wellbore drilling stability and fault gauge micro-mechanics, while the latter work was attempting to relate kerogen stiffness and anisotropy to the maturity of the Woodford shale, which is organic-rich shale. Ulm and Abousleiman [57] found "kerogen-free shale" (KFS) to be strongly transversely isotropic at the nano- and micro-scales mechanics measurements while estimating the normal and transverse Young's moduli. These early nanoindentation studies [57, 58] were attempts to measure shale mechanics at the smallest possible "porous unit" of a mudstone rock, i.e., attempting to identify the scale of the representative elementary volume (REV) [19], of fluid-filled shale composites. Their shale samples used in these early experiments contained only "trace" levels of organic material [28], which means the organic matter had little effect on the overall mechanical response (the total clay content was more than 75 wt%).

More recently, an extensive campaign of nanoindentation was conducted on the organic-rich Woodford shale ($\leq$30 % clay; 10–18 % kerogen) allowing for the observation of the effects that the kerogen matrix has on the overall mechanical properties of KRS, including its effects on the elastic and plastic behavior [7, 8, 17, 44, 55]. The upscaling of the poromechanical anisotropic parameters of KRS from nanoindenter testing and characterization, to macroscale rock mechanics laboratory measurements, all the way to field-scale logging tools has also been the subject of further studies [8, 10, 46]. However, it was found that the kerogen stiffness and the higher percent volume of kerogen reduced the Woodford shale mechanical parameters degree of anisotropy [46]. The kerogen isotropic stiffness characteristics in the Woodford shale were earlier on confirmed using the nanoindenter while measuring Young's moduli, in normal and perpendicular to bedding [61].

Very little light has been shed on how the KRS fails in tension (such as in hydraulic fracturing) or in compression (such as in drilling) at the micro- and nanoscales. Also, missing in our literature is how the polymer nature of kerogen and its spatial distribution within the shale mineral matrix affects the mechanical behavior of the KRS. Classical macroscale rock mechanics testing on KRS in both tension and compression has been performed with respect to deposition modes both parallel and perpendicular to the bedding planes of the Woodford shale [8, 52, 53]. However, the American Society for Testing and Materials (ASTM) and the International Society for Rock Mechanics (ISRM) standard test methods did not reveal any novelties about the failure mechanisms of the Woodford KRS [20]. Moreover, recently designed rock testing instrumentation for tiny shale samples under confining pressures (inclined direct shear testing device, IDSTD) [9] also showed results for compression and tension tests for the Woodford shale, which turned out to be very similar to the classical ISRM and ASTM methods.

In this work, new method for shale testing and characterization were thought off for manufacturing shale samples as micro-beams and micro-pillars, using a focus ion beam (FIB). The testing techniques and geometries utilized in this study, for the Woodford shale samples, have been the subject of material science for more than two decades when characterizing materials, such as measuring fracture toughness and tensile strength [35, 36, 43]. Using these techniques and equipment, micron-sized geometries of preserved Woodford shale were manufactured via FIB in a scanning electron microscope, SEM, and then loaded to failure via nanoindentation in an SEM [1, 42]. The associated loads/forces in micro-Newtons and failures at displacements in the range of 100 s of nanometers have shown the true nature of the failure mechanisms, in compression and tension, of the composite polymer nature behavior [27], of the kerogen-rich Woodford shale. It was found that the organic phase, in the tensile loading mode, acted like a cross-linked polymer with substantial tensile strength, and a very large tensile modulus of rupture when compared to the brittle behavior of granular shale minerals. This composite material behavior is not new to our scientific community, but kerogen tensile elastic characteristics have eluded our community to date. Indeed, this type of behavior in natural material was also observed when measuring bone tensile and compressive strength, due to the presence of cross-linked polymer-type behavior of the porous collagen/fibroblast [21, 31]. The porous nature of collagen/fibroblast embedded in bones mimics the overall composite behavior in tension of the porous kerogen, spatially distributed within the mineral matrix of KRS as shown in Fig. 1. Further, it has been very well documented in the scientific literature that organic content in bio-composites augments, by orders of magnitude, the fracture energy of these materials. One example is the hard biological tissues like shells composed of hard mineral platelets, weaved within a protein matrix [30].



Porous Fibroblast/Porous Collagen PF/PC, Wikipedia

Porous Kerogen or Organic Matter/OM Chalmers et al., 2012, AAPG

Fig. 1 Fibroblast embedded in a bone cell structure juxtaposed to a porous kerogen body (OM) embedded in an organic-rich shale in SEM

A preliminary two-dimensional numerical model, using commercial software, was built in order to model the loading and displacement curve in the composite KRS of one of the micro-beams. The emphasis was not on the fracture propagation and the strain-softening potential of beams [13, 15, 16, 23, 37] but rather on the kerogen volume vis-à-vis the other minerals components, and its intrinsic characteristics at the micro-cantilever beam support, as observed in real-time inside the SEM. The two-dimensional model did capture the micro-beam load displacement curve and its corresponding modulus of toughness at rupture.

Finally, we have intentionally ignored in this first study the poroviscoelastic characteristics of shale [4, 6, 33], as well as the poroviscoelastic and viscoplastic behavior of the rubber-like kerogen and the distributed KRS network [14, 41], which should be an important aspect of fracture healing processes.

2 The nanogranular nature of shale and its polymer kerogen

All shale source rocks have major components of non-clay minerals like quartz, feldspar and pyrites, (QFP) clays such as illite, mica, and smectite, and finally organic matter such as kerogen and bitumen where the generated oil and gas reside [38]. An unconventional shale reservoir with 5–6 wt% kerogen ($\sim$10–12 vol%) is considered kerogen-rich [48]. We will not entangle ourselves in this preliminary work in differentiating between the various types of organic matter, and we will call it kerogen. In these nanomechanics and micro-mechanics approaches, we were able to isolate the contribution of each component of the KRS composition and tie it to the different failure mechanisms observed. We may not be able to model each of these components contribution to the modes of failures independently or even their individual effects on the samples failure in tension or compression. However, we will interpret the experimental results and provide a preliminary numerical model based on the likely percent weight that the interlaced polymer (kerogen) contributes to the overall shale sample behavior at failure. The linear elastic behavior of the shale micro-beams gave no indication whatsoever from physics or mechanics to what mode of failures this shale material will exhibit. So in this study, we clearly differentiate between the linear elastic behavior of the shale micro-beams, and its parameters (Young's modulus and Poisson's ratio), and their associated mechanisms at failures.

2.1 Nanoindentation on "kerogen-free shale" (KFS): an intrinsic transverse isotropic granular material

Nanoindentation has been used to test small shale samples with only a "trace" of kerogen present, where the volume percent is assumed too small to alter the mechanical behavior of the shale at any scale. These shale samples studied contained 75–80 wt% clay [28, 57]. The shale samples were tested both parallel and perpendicular to their bedding plane with thousands of load versus displacement curves collected, which led to the characterization of the nanoscale material anisotropy in just a "trace" of organic shale. These observations concluded that the tested KFS,

with their in-bedding and perpendicular to bedding SEM images, shown in Fig. 2, are granular in nature and their anisotropic nano- and micro-mechanical properties depend on their porosity, packing densities, and the various stiffness of their corresponding minerals.

Not surprisingly, the KFS properties varied from one sample to the next, and the clay and QFP compositions varied along with their respective porosities. The SEM images in Fig. 2 were nanoindented, in-bedding and perpendicular to bedding, and exhibited clear mechanical anisotropy at these scales [57], while the organic matter content is almost nonexisting. These micro-mechanics responses and measurements confirmed the anisotropic nature of KFS at the micro- and macroscales [34].

2.2 The intertwined view of kerogen-rich shale (KRS) as a transverse isotropic composite: Woodford shale

Shale anisotropy has been well known and documented in our mechanistic approaches, in late 1960s [49], as a dry or fluid-saturated porous media exhibiting transverse isotropy. These TI characteristics are likely due to mode of deposition, bedding planes, micro-fractures and/or micro- and nanoclay grain shape and packing porosity as described in the above section. The findings of the mechanistic TI nature of shale were also modeled analytically and implemented for many field applications [5, 11, 26]. Experimental results, particularly acoustic measurements,

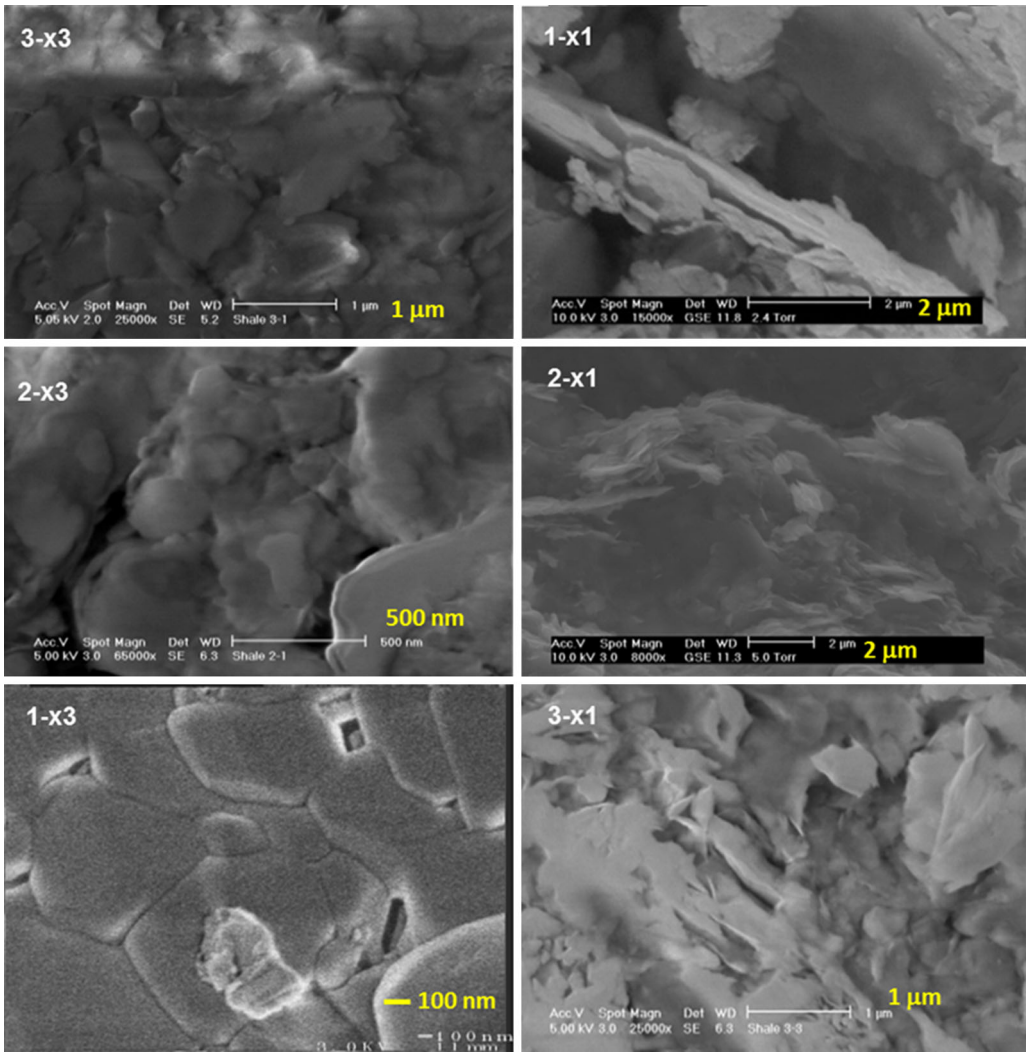


Fig. 2 Nano/micro-fabric of nanoindented KFS: SEM images of three KFS (labeled 1, 2, and 3), viewed normal to the bedding plane ($-x3$) and perpendicular to it ($-x1$). Visible in the images are the sub-micron clay particles 10–100 nm thick, which come in a large variety of forms and shapes, ranging from sheet packages (Shale 1), to wavy flake structures (Shale 2), and highly pressed and crushed sheets (Shale 3) (modified from [57])

provided the early evidence of organic shale transverse anisotropy [49]. However, later studies on source rock KRS, with the acoustic measurements, have attributed shale anisotropy not only to fractures and bedding planes but also to the presence of kerogen interlayered with illite clay minerals as shown in Fig. 3 [59]. Podio et al. [49] and Vernik and Nur [59], among many others, have paved the way for geomechanics anisotropy modeling of shale in wellbore stability analysis [2, 26], reservoir compaction simulation [5], hydraulic fracturing [11], and tiny shale samples laboratory testing characterization [9]. However, kerogen could not be definitively pinned as the culprit for anisotropy at all scales. KFS has shown intrinsic anisotropy, and in many instances even higher than KRS anisotropy, at micro- and macroscales [45, 46] and reconfirmed recently [12].

However, when the conceived mechanical structure of clay and kerogen (combined as shown in Fig. 3) is brought to failure by tensile or compressive forces, it will be extremely hard to imagine, let alone to model, the various phases and how they will interact with the rest of the shale matrix. Kerogen as a polymer has intrinsic mechanical properties for elastic behavior [46, 61] and its own material properties at plastic yield. Recent study [16] has addressed the limitation of an isotropic plastic model to be able to predict the plastic yield from nanoindentation of the KRS. The anisotropic stiffness parameters and the nature of the organic-free and or organic-rich shale and their intrinsic transverse isotropy from nano- to macroscales have also been addressed in detail elsewhere [12, 45, 47].

The Woodford shale formation, deposited during the lower Mississippian and upper Devonian period in an anaerobic marine environment, is found throughout the central part of the U.S. Midwest. The formation has long been known to be one of the major source rocks of the region, and for the past decade, it has been a great source of energy in gas and oil. The Woodford shale has high quartz content as revealed by X-ray diffraction (XRD) analysis, greater than 20 % in total porosity, and permeability ranging from 40 to 80 nano-Darcy. While it is typical of source rock shale to have kerogen dispersed in its structures, the Woodford shows pronounced intertwined kerogen strings shown in two dimensions when compared to the overall granular mineral matrix [48]. Figure 4 shows SEM of a complex and relatively very large bodies of organic material (kerogen), compared to the clay and non-clay granular particles that are embedded in the mineral matrices of the Woodford [7].

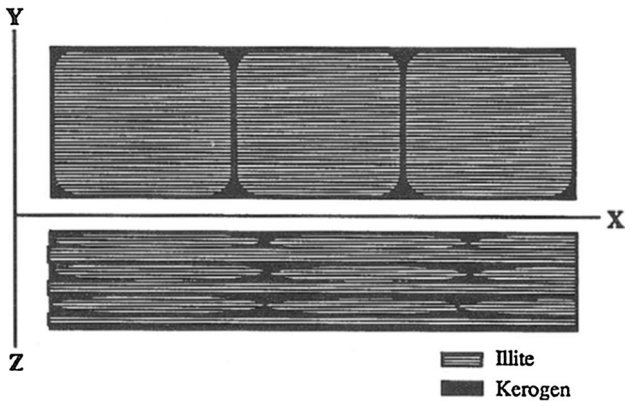


Fig. 3 Schematic of the complex layering of illite and kerogen in KRS from Vernik and Nur [59]

The heterogeneity of the Woodford KRS, like all source shale, is due among many reasons to local non-clay minerals such as quartz, calcite, and pyrite and clay minerals intertwined with kerogen string-shaped components at nano-, micro- and macrolevels. Similar to the multiscale structure of KFS [57], a complementary KRS multiscale mechanistic structure, based on SEM images, is here shown in Fig. 5. In compiling this micro- to macrostructure with micro-bedding planes and micro-fractures shown at level II, the failure mechanisms of such composite are very complex. For example, in tensile loadings, the polymer and rubber-like kerogen [12] embedded in the shale matrix, at all scales, will augment the tensile rupture (modulus of toughness) of the granular fractured structure matrix.

In the following section, we partly revisit the macroscale testing conducted on the same Woodford from the earlier studies [7, 11] for many details previously missed, since kerogen content and the composite nature of KRS were not considered in our data interpretations. In these studies, only the classical geomechanics approaches and methods of measurements were considered, for the measurement of their corresponding mechanical parameters. Figure 6 shows the loading and unloading up to failure of an ASTM 2″ in diameter and 4″ in length of a Woodford KRS sample, loaded under a confined compressive configuration. The axial and radial stress/strain curves show the slightly plastic yield deformation starting at the third round of loading and unloading (the straight dotted line on the axial deformation) and was eventually masked by piece wise partly linear slope to eventually undergo brittle failure. Yet the small strain Young's moduli shown in Table 1, at the third and fourth cycles were unaffected by the stress yield. This bilinear elastic behavior, followed by a brittle failure (not shown on this plot), is intriguing, and it is difficult to explain, considering single phase granular material nature behavior. Another observation is that the Young's moduli measured at loading/unloading cycles were more than 50 % larger than the overall Young's modulus of the full testing load range shown in Table 2. Notice that the value of the dynamic Young's moduli calculated from the compressional and shear wave

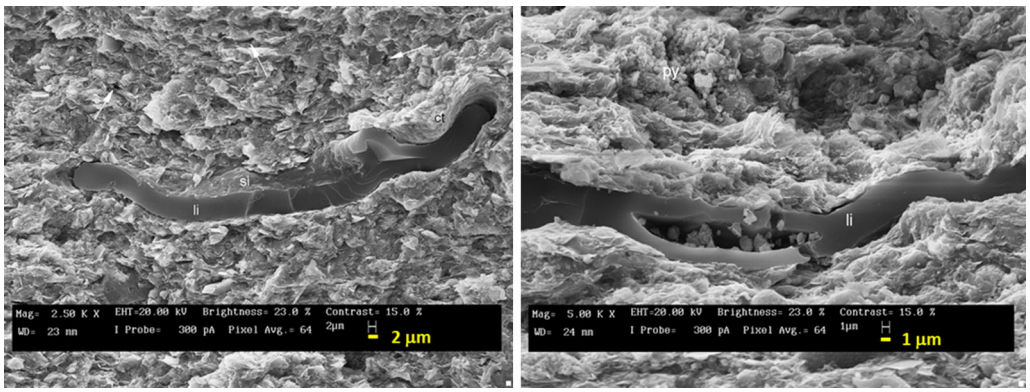


Fig. 4 SEM images of Woodford shale highlighting the intertwined nature of minerals and kerogen (*black* polymer look like *ribbons*). Notice the *scale* of these strings in 10 s of µm

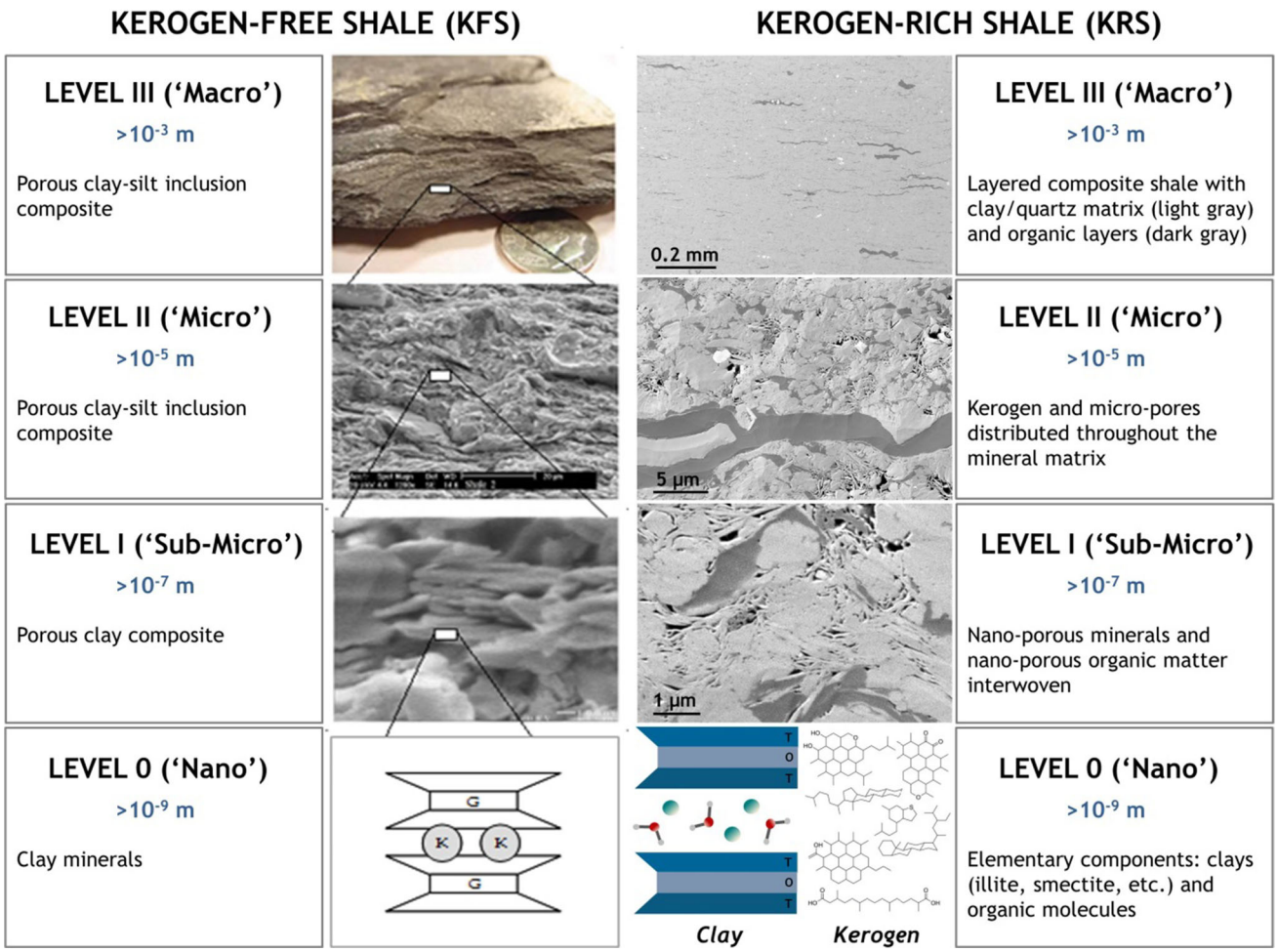


Fig. 5 Modified after [57]; Level III—the lenticular *dark colors* kerogen; Level II—kerogen occupies 15–20 % of the surface area; Level I—kerogen and matrix intertwined, quartz grains, pyrites, micro-fractures network; Level 0—atomistic *scale*

velocities was 10–15 % different from the loading/unloading small strain cycles, thus confirming the granular porous nature behavior of this shale when undergoing compressive small strain variations [7].

Nanoindentation measurements on the Woodford shale resulted in estimating transverse anisotropic mechanical parameters [57]. Table 3 shows the summary results from >20,000 indents on small, cubic Woodford shale samples.

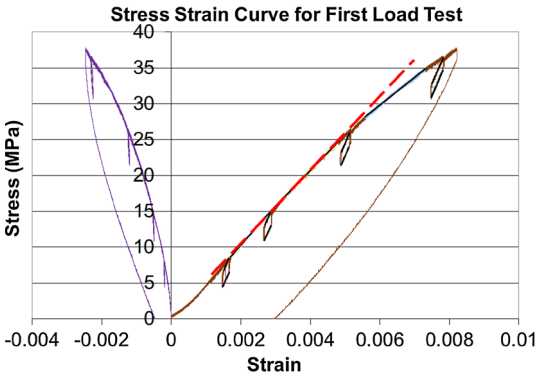


Fig. 6 Plot of stress versus strain and load/unloading small strain for Young's moduli, after [7]

Table 1 Young's modulus from small strain measurements for Woodford shale sample

Axial stress (MPa)	Cycling E (GPa)
8.6	10.8
15.9	10.5
25.5	11.1
38.0	11.1

Table 2 ASTM-recommended measurements

Sample 166-2–166-6	
Strength (MPa)	124
E (GPa)	4.5
v	0.3

Table 3 Hardness and indentation moduli of Woodford shale samples

Sample	Hardness (MPa)		Indentation moduli (GPa)	
	H_1	H_3	M_1	M_3
131′–0″	668	723	17.0	14.0
155′–0″	768	1168	17.3	21.8
166′–0″	575	1264	17.7	22.9
175′–10″	765	693	20.6	17.0
185′–10″	485	636	16.1	15.4

Obviously these numbers in hardness and stiffness show anisotropy, however not always consistent outside the same depth retrieval shale horizons.

After reviewing the data of the study conducted by Bennett et al. [17], we may have some answers or at least an explanation, for the complex behavior of the Woodford shale as described in the above. This study [17] conducted another large campaign of nanoindentation on these same horizons of the Woodford KRS [7]. In Fig. 7 is shown an example of the fidelity of the equipment used, evident by the clarity of the various minerals and non-mineral phases and minerals that were identified at these micro-scales. Clearly, frame (b) in Fig. 7 shows that under the indenter imprint the organic matter, clay and non-clay minerals were all loaded simultaneously. The full sweep of tests on these samples both parallel and perpendicular to bedding showed that the organic matter (OM) has an anisotropic stiffness, yet much smaller in value than the ones reported by earlier study [61], in the plane parallel to beddings. However, it was indicated in their discussion section [61] that "damage may have had occurred during cutting and polishing, due to frictional heat, thus altering the inherent kerogen anisotropy."

3 Experimental methods

Focused ion beam and scanning electron microscope, FIB–SEM, were used for sample preparation of specific geometries such as the micro-pillars and the micro-

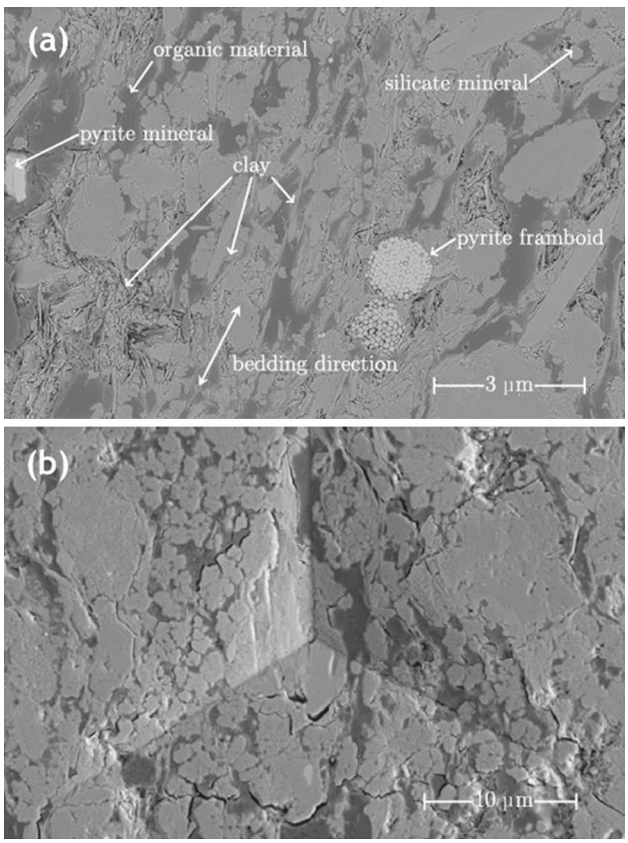


Fig. 7 Woodford shale SEM images after [17] showing **a** a polished surface with organic material which includes 1-μm-diameter pyrite framboids, silicate, clays, etc., and **b** a similar region which has been indented

cantilevers of the Woodford KRS. Four micro-beams and three micro-pillars were milled and prepared for testing. However, there are only five successful test results, to report in this study, the four micro-beams and one micro-pillar (the other two micro-pillars collapsed when testing next to the first one, Fig. 9a).

3.1 KRS micro-beams using a picoindenter (PI-85) in the FIB–SEM

A sample with dimensions of 1 × 1 × 0.4 cm was cut from a preserved Woodford KRS cored well [7]. A sharp 90° edge was created by mechanical polishing using standard silicon carbide paper up to 4000 grit followed by polishing, with 1 µm diamond grit. A Quanta 3D field emission gun (FEG) with FIB–SEM was used to prepare the micro-beams shown in Fig. 8b. FIB surface milling was used to clean the surface for better sample imaging as well as to prepare the desired micro-beams geometries. Four micro-beams were successfully manufactured, using the FIB procedure according to the S.G. Roberts method [29, 43]. Each shale micro-beam was shaped by cutting trenches on all three sides with widths of 20 µm and depths of 10 µm using a 15-nA beam current, resulting in a *U*-shaped trench. The geometry was then refined by applying a 1-nA beam current. Afterward, the sample was tilted to 45° along the length axis to shape the micro-cantilever. The base of the micro-cantilever was undercut from both sides using a 3-nA beam current. The resulting micro-cantilever geometry is shown schematically in Fig. 8a, with the corresponding SEM images of three of the four micro-cantilever beams tested as shown in Fig. 8b.

A Hysitron Pi-85 picoindenter used to load the micro-beams under displacement control mode, at a rate 15 nm/s. The indenter tip has a flat circular punch geometry, with a diameter of 5 µm. All loading experiments were performed inside the SEM, where loading of the micro-cantilever beams continued until failure. The indenter tip was placed at the end of the beam, centered along the *y*-axis as shown in the SEM in Fig. 8c. Load and displacement data were collected in real time.

In Sect. 2.2, we illustrated from macromeasurements on 2″ diameter × 4″ length samples that the loading/unloading Young's moduli differed from the large strain Young's modulus by more than 50 % but are within 15 % of the dynamic measurements. Also, it is well known by now that the values of Young's moduli obtained by nanoindentation on porous multiphase material are close in value to the small strain deformation and to the ultrasonic pulse transmission velocity measurements [57]. However, when a

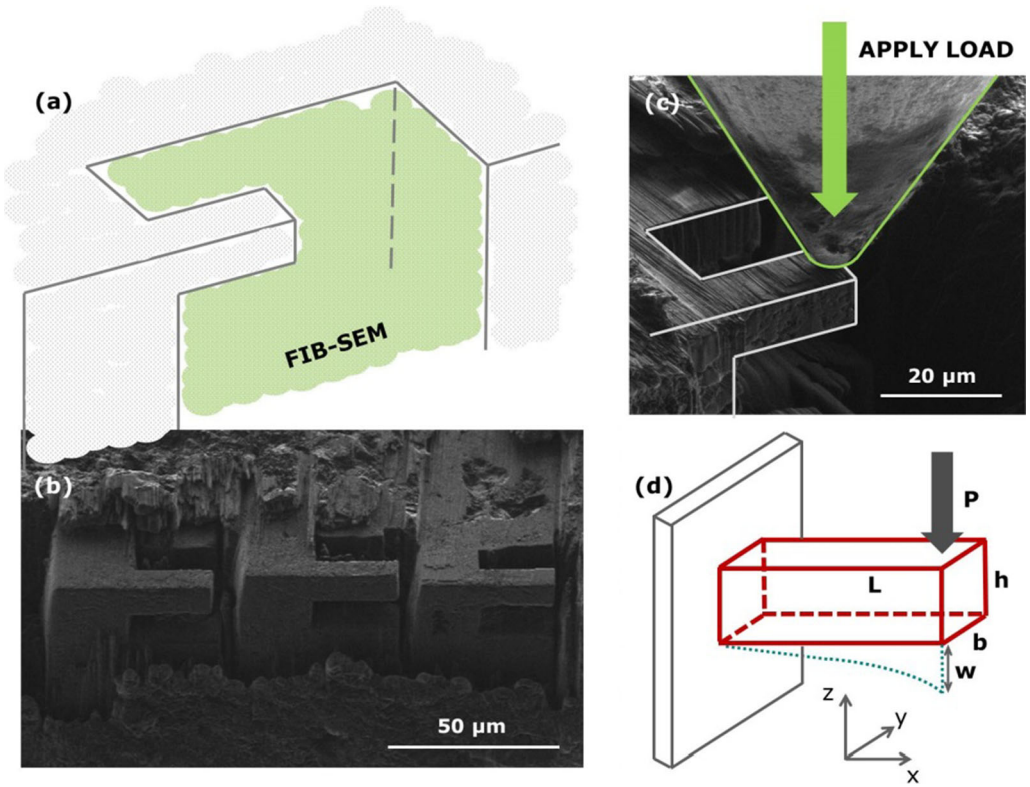


Fig. 8 **a** Schematic of micro-cantilever beam manufacturing is shown, **b** an SEM image of three cantilever beams, **c** SEM image of the first micro-cantilever beam at loading and **d** schematic of the micro-experiment with the dimensions

solid metallic beam with micron-sized dimensions is subjected to loading, there is strong evidence that size effects come into play. Chen and Feng [22] have elaborated on this phenomenon and have obtained the theoretical results corresponding to an intrinsic length scale effects on the overall micro-beam deflection, w [shown in Fig. 8 frame (d)], of a solid micro-cantilever beam with intrinsic length scale, l_{FE}, that is found by calibrating a typical beam thickness with the experimental suite of results. Although they have derived expressions for micro-cantilever beams subjected to various loading conditions, we only include the expression for the one relevant to our experiment (concentrated load on a micro-cantilever beam) as in Eq. (1):

$$W = \frac{Px^2(3L - x)}{6E(I + bhl_{FE}^2)} \tag{1}$$

where I is the moment of inertia, for the micro-cantilever beam prismatic cross section, b and h are the cross section dimensions, and L is the micro-beam length shown in Fig. 8d. The above expression provided satisfactory results when used to analyze the experimental measurements by Lam et al. [40]. The classical theoretical expression of the micro-cantilever beam was more than 300 % off from the experimental results yet monotonically improving as the thickness, h, of the beam reached the 100-s micrometers level. In our case, for a granular multiporous structured material intertwined with organic matter, we are not sure what the above expression should look like or even where to start in formulating the corresponding phenomenological theories. The discovery, however, is that when we assumed $l_{FE} = 0$ from Chen and Feng [22] in Eq. (1), where it turns into the expression for the classical theory of beams as in Eq. (2) [56], the micro-cantilever beam Young's moduli were within 10 % error from the ones shown in Table 1 and the published values in [17]. Indeed, results calculated from the classical theory of beams for the stiffness expression from Eq. (2) match the nanoindenter results as well as the small strain loading/unloading of Fig. 6, which are also the corresponding values, to the ultrasonic pulse transmission velocity measurements when using compressional and shear wave velocities [7].

$$E = \frac{PL^3}{3wI} \tag{2}$$

3.2 KRS micro-pillars using a picoindenter (PI-85) in the FIB–SEM

Similarly, prismatic micro-pillars, with minimal taper, were manufactured in the FIB instrument (Quanta 3D FEG) with successively lower beam currents (5 nA down to 0.3 nA at 30 kV) to achieve the micro-geometries shown in Fig. 9. The milling procedure followed the methods of earlier work used in material sciences [35, 36, 43, 51]. To achieve the square geometry, the samples were tilted by ±2° with respect to the incident ion beam in order to mill the side surfaces of the pillar by grazing incident ions. The aspect ratio (micro-pillar height divided by width) was set close to 3:1 as in earlier studies [1, 35]. These dimensions are not perfect for the shale samples shown in Fig. 9a and may vary slightly; eventually, if these tests are to be adopted and standardized for geomechanics and rock mechanics, much more testing results and test analyses will be needed.

A different Hysitron indenter was used to uniaxially compress the micro-pillars using a diamond flat punch tip indenter, with 60° conical and a 10-μm-diameter flat tip, as shown in Fig. 9c. The micro-pillars were loaded at a predetermined displacement rate of 15 nm/s until failure. Figure 9b provides a schematic of the micro-pillar with dimensions of b, h, and L and the applied force P. The compression loads of the micro-pillar samples, under the nanoindenter, mimic the classical macroscale unconfined compressional tests, for stress–strain relationship and the unconfined compressive strength (UCS measurements).

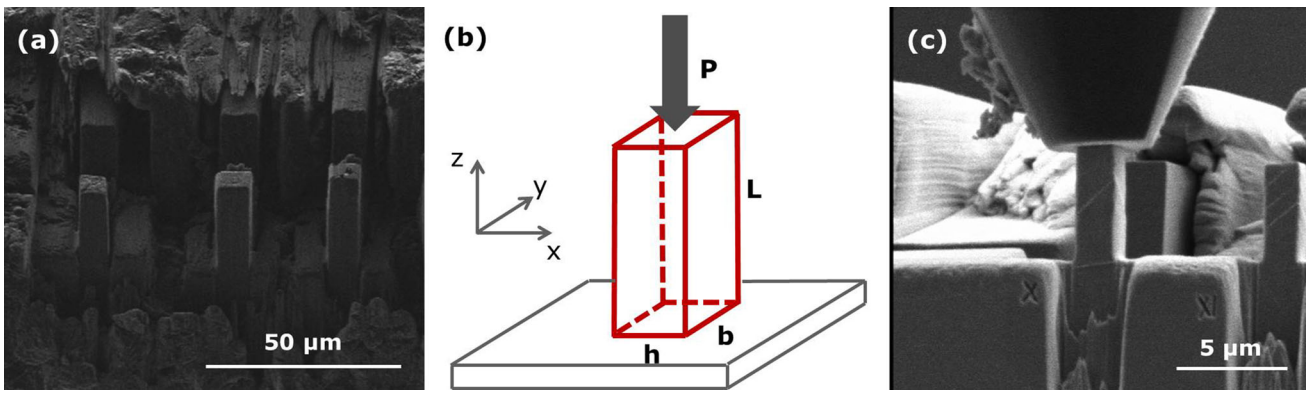


Fig. 9 **a** SEM image of the three-pillar pre-test, where only the left pillar was tested successfully, the other two failed on the impact of the first micro-pillar shear failure, **b** schematic of micro-pillar loading and **c** SEM of a micro-pillar while loading

4 Results

4.1 Micro-beam testing

The four micron-sized beams milled for this initial attempt, to load and fail the Woodford shale samples in tensile mode. Given the heterogeneous nature of all shale, we were interested to see whether at the micro-scale we can reproduce the mechanical response from one test to the next. So, each test was performed inside the SEM with a small-scale nanoindenter, and movies of the loading and failure were captured in real time during the experiments. This unique setup provided us not only the ability to load and fracture micro-scale KRS structures, but also the advantage of visualizing the initiation of a fracture in the micro-cantilever beam tensile zone and then propagation and ultimate failure. These observations were captured in real-time video, while correlating these phenomena step-by-step to the force–displacement plots were also collected during the experiments. Figure 10 shows the four micro-cantilever beam load–displacement curves. T3 and T4 failed in brittle failure modes, while T1 and T2 showed large ductile/plastic deformations before failure. We will interpret, in the followings, each micro-cantilever beam test independently.

4.2 Test 1: Linear elastic behavior in pre-yield and strain-softening in post-yield; last an elastic recovery

Figure 11 shows four stages of test one, T1, with the load–displacement correlating with the real-time SEM pictures (snap shots from the real-time video) of the micro-cantilever beam subject to constant displacement/rate loading configuration, 15 nm/s, up to the sample failure. In Fig. 11a, the beam has been continuously loaded up to a load, $P = 809$ μN, with a displacement, $w_1 = 697$ nm, in a linear elastic load deformation curve. A sudden drop in stress occurs, and a tiny transversal crack shows along the micro-beam fixed support, bringing the displacement $w = 2423$ nm. However, the beam continues to deflect and soften, with a complex strain-softening post-yield behavior, while the fracture continues to propagate toward the bottom of the micro-beam as shown in frame (c). In this frame, it is observed that the micro-cantilever beam has eventually totally failed, and it is "almost" separated from its fixed support with deflection, $w_3 = 4499$ nm.

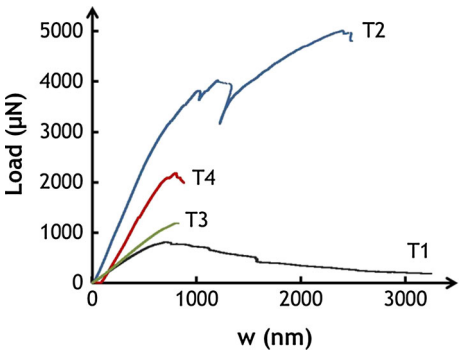


Fig. 10 Four micro-cantilever beam loading/displacement *curves* plotted in the above, T1 and T2 show strain-softening and strain-hardening behavior, respectively, before ductile/plastic failures, while T3 and T4 show brittle failure with little or no yield

Finally, in frame (d) a rebound of the micro-beam is observed from frame (c) to a final deformation position, when the indenter is totally lifted, leaving a residual deflection, w_4. Notice that $w_4 < w_3$ as shown in the dotted lines, evidenced by the displacement recovery from frame (c) to frame (d) displayed also on the load/displacement curve. Interpreting this behavior is extremely interesting, backed by real-time observation. In Fig. 11 from frames (b)–(d), a polymer-based rod-like object stretching and yet still holding the failed micro-beam to the severed support is observed. In Fig. 12 is displayed the cut and paste of Fig. 11 frame (d) juxtaposed to an SEM of the same Woodford shale horizon, showing the lenticular kerogen, and the full load/displacement curve with four steps (1–4) marked on the strain-softening part. Notice the scale and shape of the lenticular kerogen, just like strings shaped, weaved within the shale mineral matrix. Also, looking at the scale in Fig. 12b, these strings can reach more than 100 μm in length and 5–10 μm in thickness (not accounting for the third dimension). It is easily imagined that this polymer-like kerogen can be embedded in the total length of the micro-beam and way further into the micro-cantilever beam fixed support. This composite mineral/organic structure is responsible for the micro-beam displacement recovery from steps 3 to 4 as shown in Fig. 12c and as observed in real time. (See also the "Appendix" for a zoom out for more details of this very interesting behavior.)

In contrast to granular material failure, obviously we can notice that the polymer-like string keeps the beam attached to the support after a total tensile failure of the micro-beam and is responsible for the strain recovery. The shale matrix granular nature is broken as shown in Fig. 13, yet the micro-cantilever beam is still hanging on to the support after the nanoindenter was unloaded. This behavior is typical of macrocomposite beams such as reinforced concrete beam [15, 16] or in geogrid-reinforced site constructions [18]. Post-failure analysis shows strain-softening behavior that can be easily reproduced using numerical simulation. Since the kerogen and other organic content in this Woodford KRS was never observed in tensile loading or tensile failure to have such a drastic effects on the mechanical behavior. The constitutive model for the mechanical behavior of the kerogen matrix intertwined

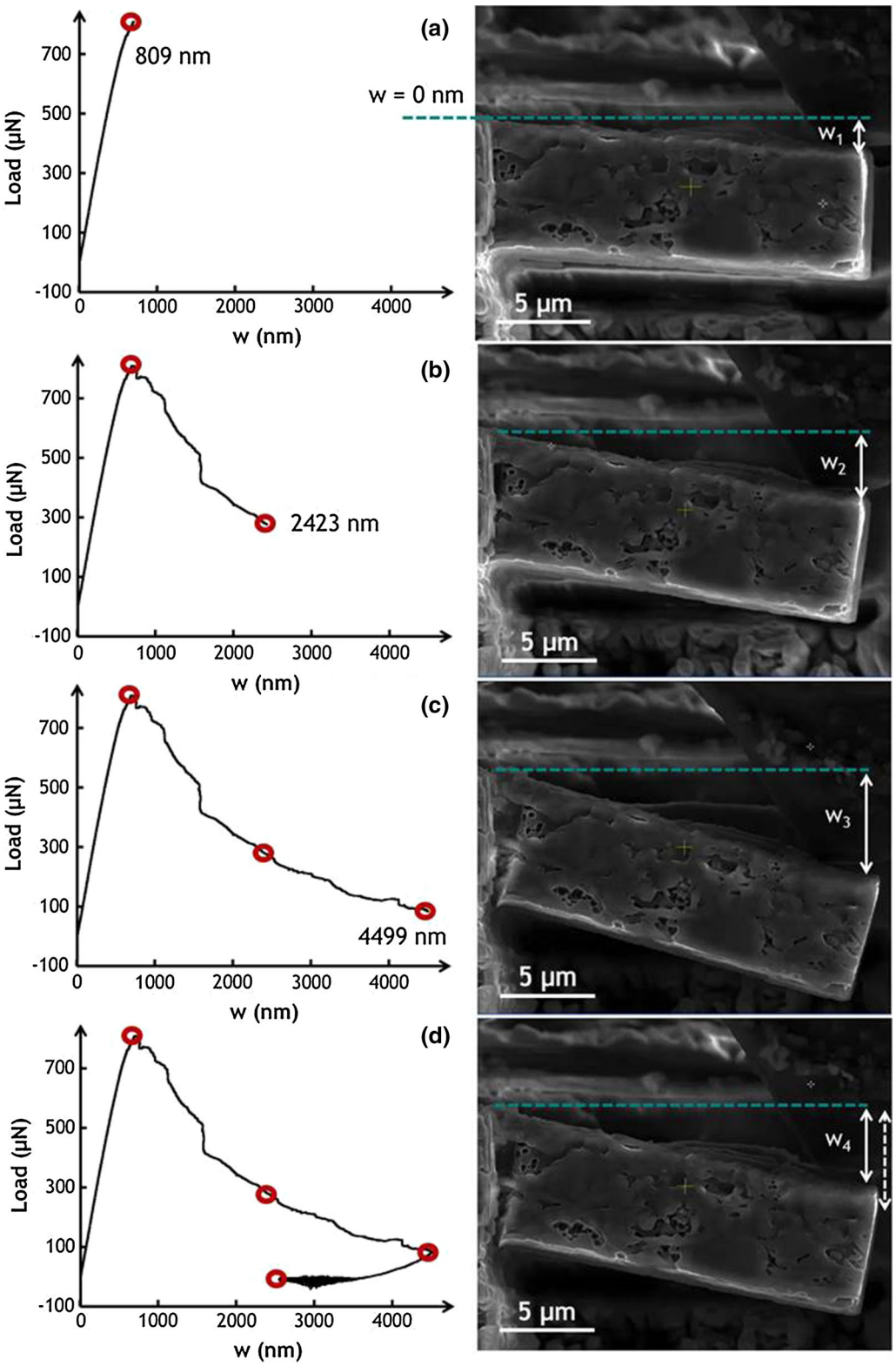


Fig. 11 Frames and plots captured from micro-cantilever beam loading, T1, displayed above

with other shale minerals is nonexistent. To feel comfortable with our explanation of the observed strain-softening behavior of the micro-beam, we built a simple two-dimensional numerical model mimicking the micro-beam response in this one test. This is an attempt to explore the potential constitutive model for the micro-cantilever beam

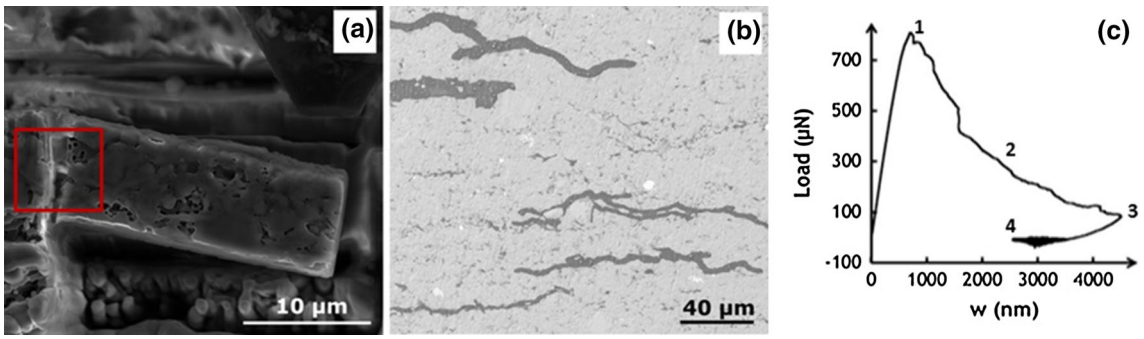


Fig. 12 **a** SEM image of T1 at failure with a *red square* highlighting the organic rod-like material, **b** SEM image of Woodford shale from the same horizon with the interlaced lenticular kerogen in *dark colors* and **c** the full load–displacement curve

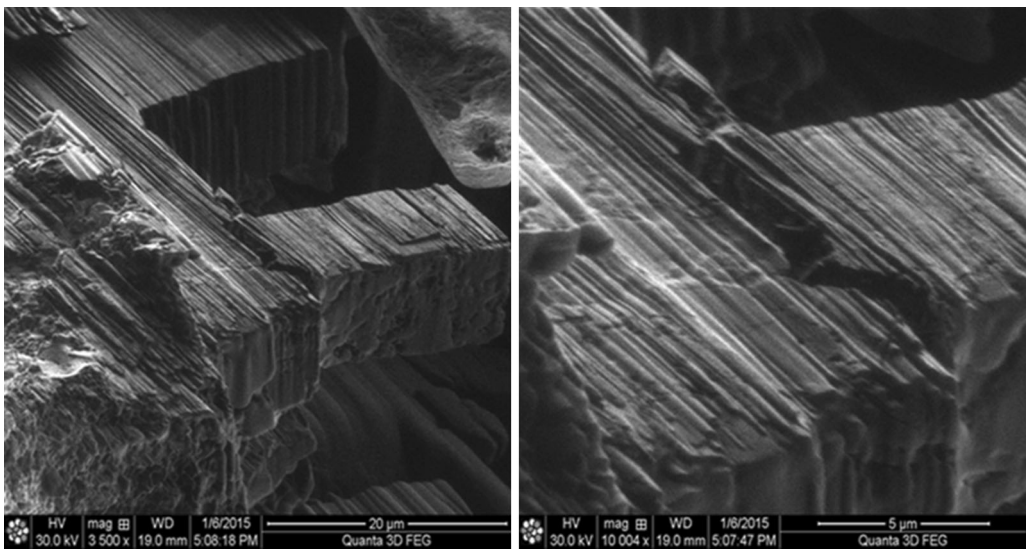


Fig. 13 SEM image of T1 a total collapse and the severed plane of failure while unloading the indenter

mechanical behavior, through matching the force–displacement curve by placing a percent of volume of kerogen with sustainable tensile strength characteristics at the support.

4.3 Test 1: Numerical modeling of the micro-cantilever beam behavior

SEM images indicate that the kerogen content is quite rich at the fixed end in Test 1 (T1) and thus contributes to the large bending deformation of the micro-cantilever beam. In this initial plane–stress numerical model, the real dimensions geometry of the micro-cantilever beam in T1 was used very closely, i.e., 22 μm in length and 9.0 μm in thickness. The out-of-plane direction is unity (1 μm), but the loading force monitored in the numerical model was scaled by 9, to account for the thickness of the micro-cantilever beam in the nanoindentation test configuration. A quadrilateral element was used in FLAC, with uniform elastic material constants such as Poisson ratio of 0.25 and a Young's modulus 14 GPa.

From the experiment results and observations, most of the tensile yielding took place at the fixed support, consistent with the larger moment force for a cantilever beam behavior. To simplify the setup in the numerical model, the left column of elements ("kerogen") is assigned with strain-softening capability, while the rest of the elements are modeled as pure linear elastic material. In the schematic model, shown in Fig. 14a, the left boundary was fixed in both *X*- and *Y*-directions. The grid point at the upper right corner was loaded in the downward direction at a constant rate of 1×10^{-5} μm/step. The reaction force and deflection at the loaded grid point were monitored during the entire course of the simulation. The modeling approach was to adjust material properties so that the force–deflection curve modeled in the numerical simulation matches with the measurement in the laboratory test shown in Fig. 11. The elastic response (before the peak)

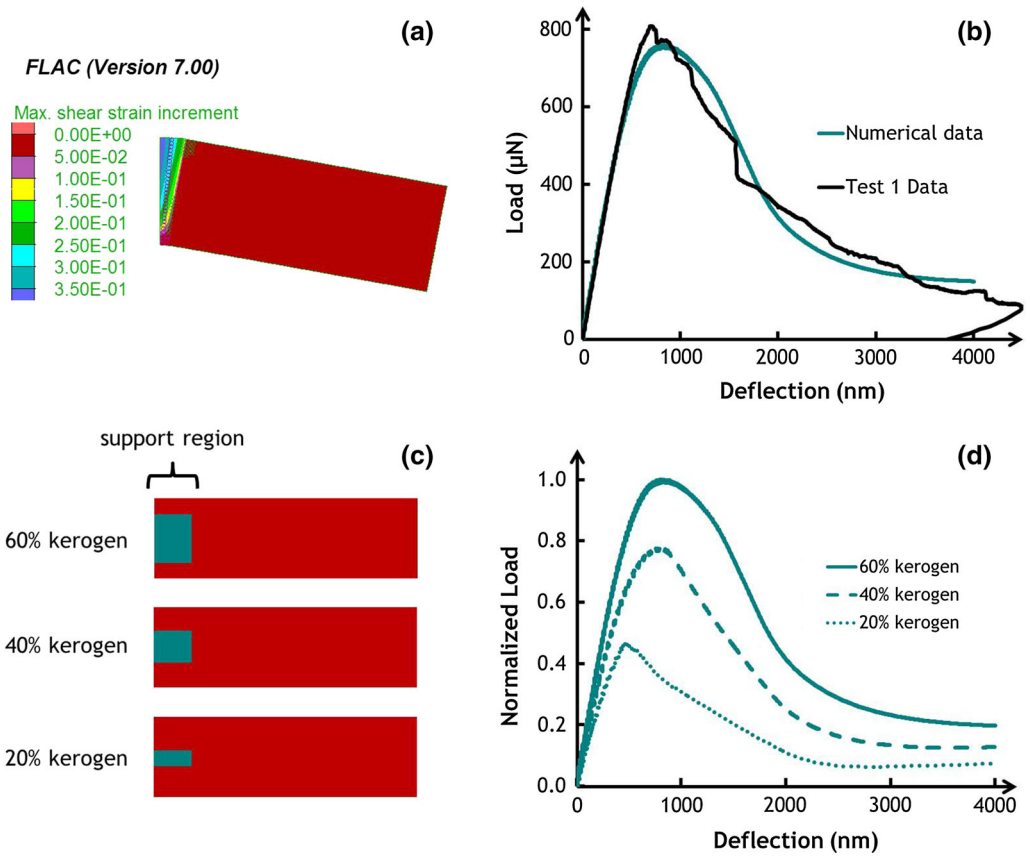


Fig. 14 FLAC modeling of micro-beam T1: **a** the contour of maximum shear strain with tensile yielding indicator, **b** the comparison of force–deflection curve measured in the laboratory test and numerical model. Sensitivity of bending strength of micro-beam to kerogen content, **c** three simulation cases with kerogen content varying from 60 to 20 %, and **d** normalized force–deflection curves from three cases also illustrating the modulus of toughness variation

was matched by adjusting material stiffness in the model. The plastic component was also matched by adjusting the strain-softening curve of the kerogen material. The simulation indicated that a Young's modulus of 14 GPa in tension and a bilinear tensile strain-softening seemed to give quite a good match, as presented in Fig. 14b. That is, the tensile strength was 130 MPa initially, decreased to 85 MPa at plastic tensile strain of 17 %, and further dropped to the residual tensile strength of 12 MPa at plastic tensile strain of 70 %. The tensile crack area developed near the fixed end in Fig. 14a also looked similar to the laboratory observation. Note the Young's modulus used in the numerical model is very close to the values measured for the four micro-beams reported in Table 3. (Test 2 results are different for a very good reason, as will be discussed later.)

Using the material properties calibrated in the base case (60 % kerogen), the simulation was again repeated two more times, with kerogen mass reduced to 40 % in the second run and 20 % in the third run (see Fig. 14c). The monitored load–deflection curves in three runs are compared in Fig. 14d, in which all the forces are normalized by the peak force recorded in the base case. The degradation of shale tensile strength with kerogen content was proven in this sensitivity study.

4.4 Test 2: Linear elastic pre-yield and strain-hardening post-yield before a sharp snap at failure

Test 2 (T2) was performed in two stages shown in Fig. 15. The blue curve shows the first stage of load/unload highlighting relatively elastic behavior with one minor kink. The kink observed at 3000 μN, yet the micro-beam continued displaying elastic behavior during loading up to 3500 μN. The beam then unloaded, holding at point 2 then loading and unloading again to confirm its elastic linear behavior [7]. Indeed, we obtained as seen on the blue curve, approximately an equal and parallel slope to the first loading curve and an almost total recovery of the elastic

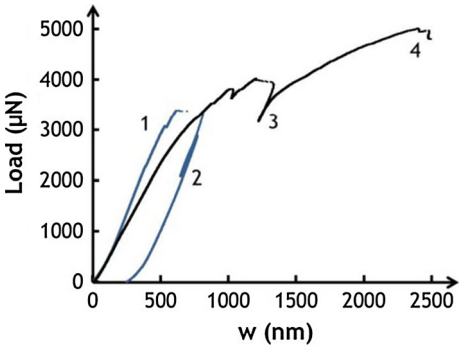


Fig. 15 Plots of the two-load stage load/unload then reload to failure, for micro-beam Test 2

linear displacement. The black curve in Fig. 15 shows the second stage, where the same micro-beam immediately reloaded, without any disturbance in between stages or even any time lapse. The progress of this loading, as recorded in real-time video, is herein video captured and illustrated in Fig. 16 from frames (a)–(d) in a chronological order. Frames (a) and (b) showed a linear elastic reloading behavior, up to almost 3000 μN. Then passing the threshold of the earlier elastic response, as shown in Fig. 14, a kink was observed suddenly at 3800 μN, and simultaneously a tiny transversal crack shows up on the beam. Then the micro-beam is recovered shortly to a load value close to 4050 μN before a substantial failure, and a major crack at the support is clearly observed on the micro-beam picture in frame (c). Meanwhile the load decreased to a little $<$3000 μN shown also in Fig. 15 as point 3. As the loading continues beyond this point, a strain-hardening behavior was observed. However, the micro-beam continues to build strain energy before the final snap and total failure at point 4, in Fig. 15, and is captured in frame (d). This ultimate tensile load of KRS is equivalent to unconfined tensile strength, UTS, which is a value that carries much significance in our geomechanics source shale field hydraulic fracking applications. In fact, the UTS in KFS could be only equivalent to the tensile strength of the rock shale, T; however, in KRS it is much more complicated than the classical tensile strength of a rock, as evident from the two micro-cantilever beam results. In particular, since hydraulic fracturing is a tensile Mode I crack opening, crack failure, and crack propagation, in a composite polymer-type weaved porous material, the mechanics may be more complex than what we have in our hydraulic fracturing models. Finally, in Fig. 15, the two superimposed stages show that the early loading of the second stage experiences no loss of elastic energy up to 1000 μN. A little overall weakening of the shale matrix structure is observed (evidenced by a smaller slope and typical of natural granular porous material), but not the overall strength of the micro-beam, as it eventually failed with a much more strain energy, stored in the system, and corresponding to a relatively very large modulus of toughness.

4.5 Tests 3 and Test 4: Linear elastic and brittle failures with minimal or no yield

Micro-cantilever beams T3 and T4 are shown in Fig. 17, where sharp brittle failures with clear snaps were observed during the testing while SEM imaging. The brittle failure is indicative that there are only trace amounts of organic matter at the fixed support of the micro-cantilever beam. As defined early on, "trace" organic matter is an amount that is not enough to alter the mechanical behavior. This fact was illustrated in the numerical simulation in Fig. 13d, where reducing the volume percent of kerogen at the support diminished the strain-softening as well as the ultimate load observed during elastic loading. Again, we notice for T3 and T4 that the Young's moduli measured at 50 % of the maximum load in the elastic regime are very close in value, as shown in Table 3, yet their failure loads varied by almost 100 %. In Fig. 18 is shown the real-time load/displacement video, captured for illustration of the micro-beam, T3, barely reaching the 800-nm displacement to snap into a brittle total collapse.

In summary, the four micro-beams showed very interesting behaviors within a span of 200 μm in the preserved sample of Woodford KRS. In Table 4, we summarize the dimensions of each micro-cantilever beam, as sketched in Fig. 8d, to illustrate the difficulty of attempting to obtain exact dimensions for each milled porous micro-beam. The calculated values of the Young's moduli were taken at $\sim$50 % from the linear elastic loading span, i.e., they were calculated based on picking up the corresponding load, P, and the deflection, w, at 50 % on the four loading curves.

Finally, the modulus of toughness, defined as the work/energy needed before the total rupture of the beam, is illustrated in Fig. 19. You can see the two shaded areas where T3 required $\sim$15 % of the toughness needed to break micro-beam T2. A similar comparison can be made between T3 and T1 although the magnitude is different. This is what we are faced with in the tensile cracking of these composite KRS formation when we are hydraulically fracking.

4.6 Micro-pillar compression testing

While three micro-pillars were prepared for testing, only one ended up being loaded successfully to failure. During loading of the first micro-pillar and at failure, the other two adjacent micro-pillars broke unexpectedly, indicating that for this shale the milled micro-pillars were too close to each other as shown in Fig. 9a. Earlier study has used EDS to analyze nanoindentation results and to isolate

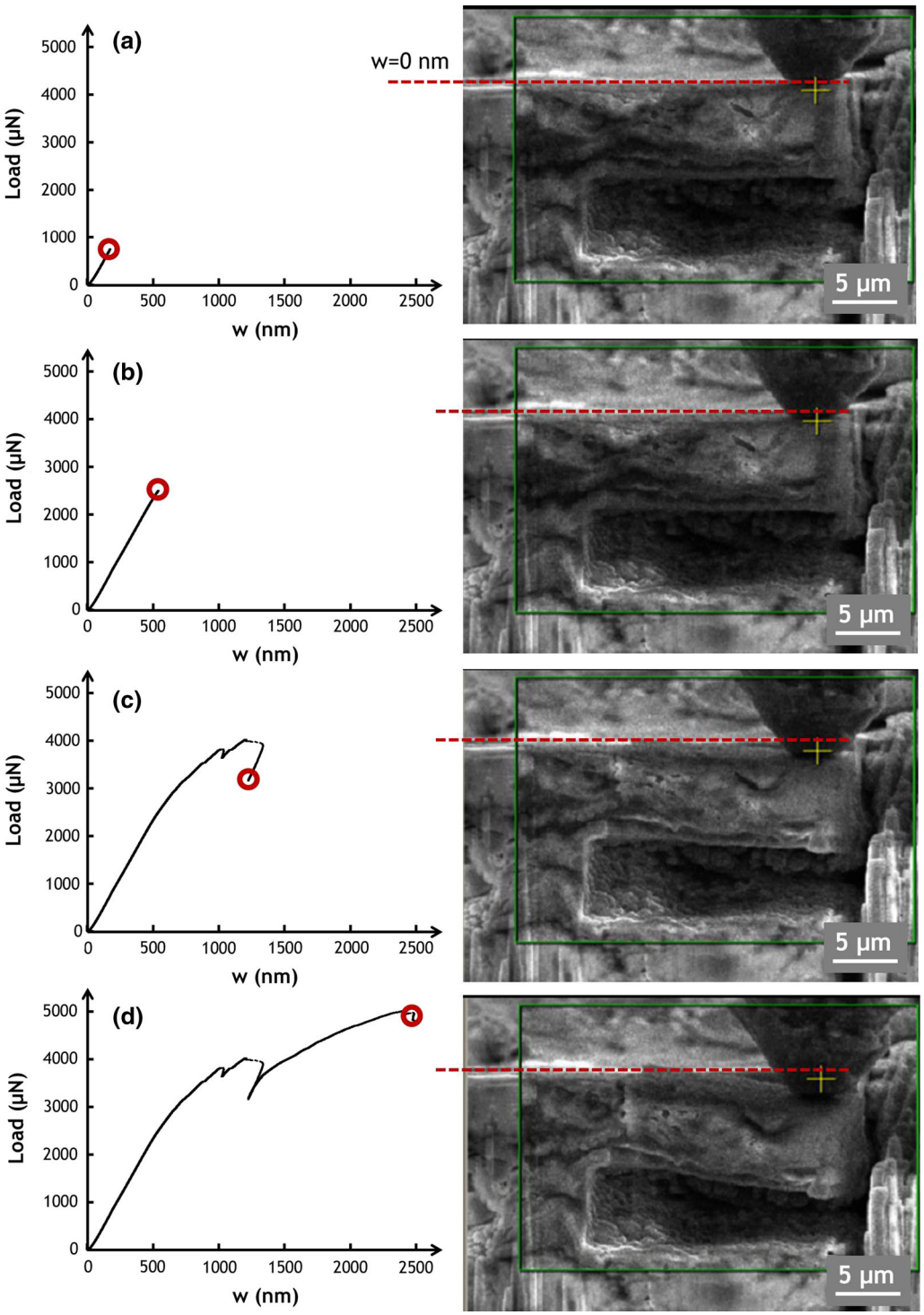


Fig. 16 Four SEM frames captured from T2 are provided in chronological order for the reloading, showing the final load/displacement progress up to point 4 in frame (**d**) as shown in Fig. 15. The fracture has already propagated as the load increased from 3500 to 4000 μN across the depth, *h*, in frame (**c**) yet the micro-beam regained strength and the load increased reaching above 5000 μN before total rupture

mechanical phases when describing Woodford mechanical behavior [25]. In this test, EDS was conducted on the micro-pillar face before testing as shown in Fig. 20a. A preexisting inclined shear band-type geometry can be observed across the center of the micro-pillar, as evidenced by the orange color, which is enlarged in the superimposed

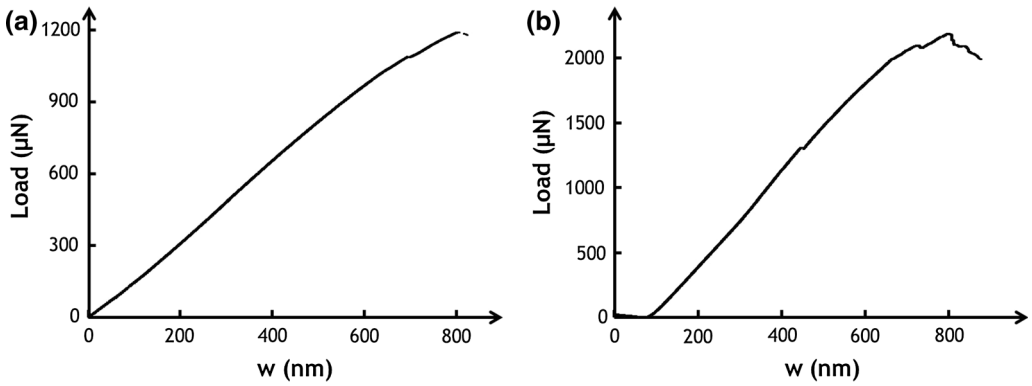


Fig. 17 Plots of the two micro-cantilever beams Test 3 (**a**) and Test 4 (**b**)

frame (b). The top of the pillar has a high concentration of silicon and oxygen, indicating silica, the middle band contains aluminum, silicon, potassium, and oxygen, indicating clay, while the bottom section has a high concentration of calcium and magnesium, indicating dolomite.

The test results in Fig. 21a show the various stages of the stress–strain curve. Initially, we observed a nonlinear loading section, and as the load increased, the micro-pillar exhibited a linear elastic behavior, illustrated on the plot with the line extension, before the micro-pillar goes into short yield and then a load–unload section and eventually failure. This stress strain curve resembles exactly, in its stages, the uniaxial 2 in. × 4 in. Woodford KRS sample in Fig. 6. Even the load and unload part of the curve shows a higher Young's modulus, consistent with the load/unload cycles in Fig. 6. Since the loading/unloading occurred after a short yield, the higher Young's modulus could be due to permanent micro-pillar consolidation or pore/grain compaction. Eventually, the shear band-type geometry acted as a weak plane along which the micro-pillar sheared during failure shown in Fig. 21b with the superimposed pre-failure EDS map in (c). This is not surprising, since shear bands often form when granular materials undergo ductile behavior under some loading configurations [24, 39]. Having this preexisting condition, the shear failure showed little interference of the kerogen polymer-like behavior.

5 Discussion

The micro-beam in T1 exhibited ductile behavior as shown earlier, and it post-yielded in a strain-softening regime, while the ductile behavior of Test 2 demonstrated strain-hardening in post-yield. Meanwhile the micro-beams in Tests 3 and 4 exhibited brittle failure modes. Determining the reasons for the differences between each of the failed micro-beams shown in Fig. 10 is important for us to be able to upscale and convert this understanding into predictive tools when it comes to hydraulic fracturing, wellbore drilling, reservoir optimal productivity, and many other oil and gas field applications. We will outline our thoughts using a mosaic of high-quality SEM images, giving the reader the opportunity to formulate their own views.

We have observed in these KRS tensile loading experiments the role of kerogen and other organic matter as cross-linked polymer [27, 38] contributing substantial tensile strength to the shale micro-beam. In T1, the beam's granular structure totally failed as early as frame (c) in Fig. 9, and yet the kerogen string shown in Fig. 12a rebounded the whole beam back, recovering some of its elastic energy ("Appendix"). In Fig. 22 frames (a) and (b), we tried to zoom in on this observation to show that, indeed, the beam had separated from its support, and yet the tensile elastic string stretched but did not rupture or pull out but rather was still fully embedded in the beam. In other words, the kerogen string never reached its full "modulus of toughness" or total energy needed for rupture, while early on in the loading range the granular part of the beam reached its "modulus of toughness" in tensile loading ("Appendix"). The granular part of the beam is illustrated in Fig. 22, with top view frame (c) and the zoom-in frame (d). The amount of kerogen at the support was enough to give it the strain-softening behavior, where the micro-beam acted as a composite material, such as reinforced concrete beams. In micro-beam test, T2, shown in Fig. 23, the amount of kerogen was large, and even overwhelming at the support, with little volume of the clay or non-clay granular material. The volume of the organic matter that stayed behind at the support is evident by the large cavity left on the micro-beam after total collapse shown in Fig. 22 frames (a)–(d). The volume of the kerogen was large enough and stiff enough to carry the micro-beam into a strain-hardening post-yield type of failure, with a huge modulus of toughness contributing to the overall

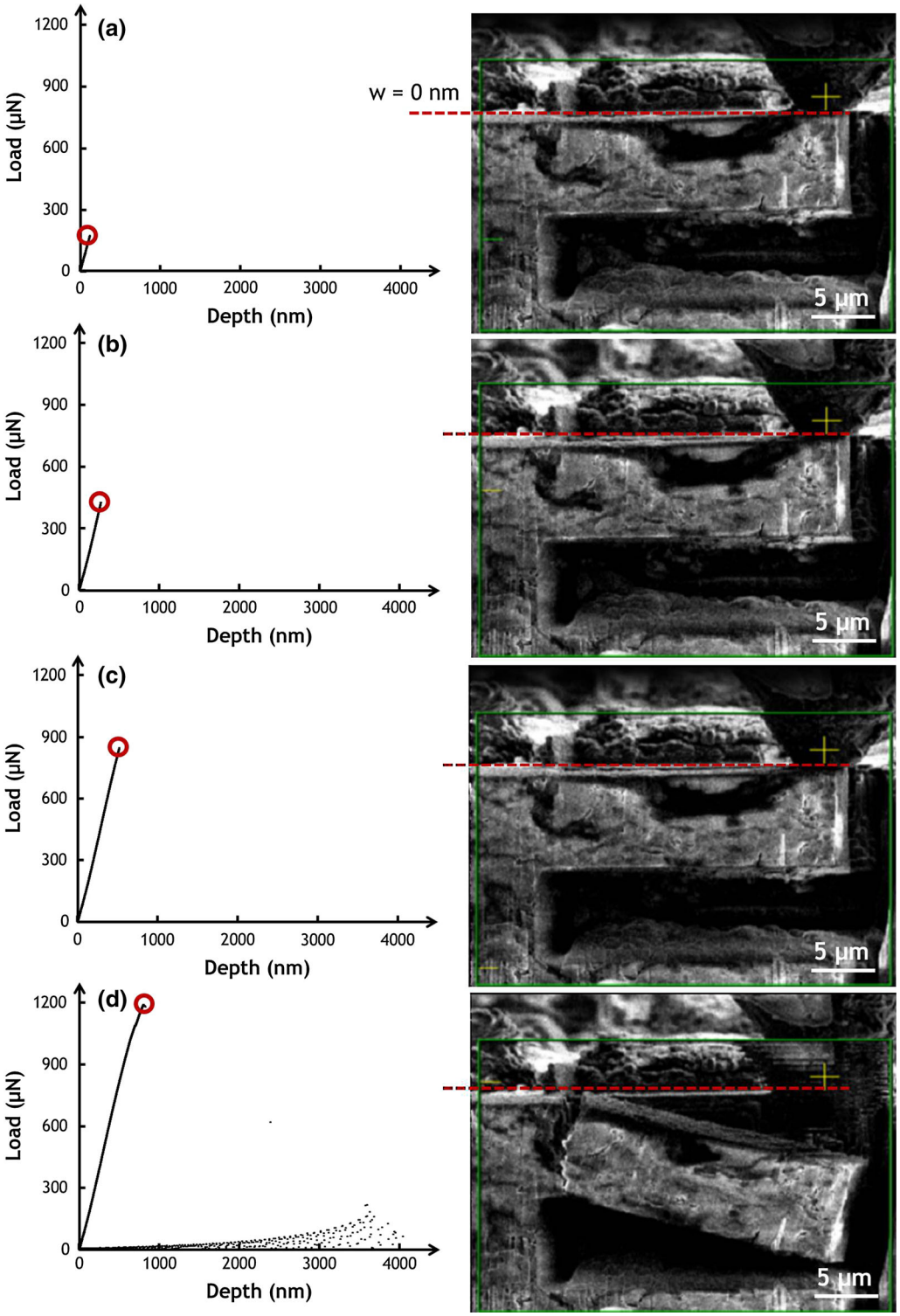


Fig. 18 T3 just before its brittle failure (*top frame*) and then just a few seconds after failure (*bottom frame*) with the indenter partially retracted

shale micro-beam behavior. The progress of failure in T2 reinforced our early hypothesis that the cross-linked polymer nature of kerogen and its intertwined structure with the non-clay and clay mineral matrix is the one holding the granular shale matrix together, resulting in large and unexpected values for granular material in tensile failures. Also from Fig. 15, observed in frames (c) and (d), the fracture has totally developed across the depth of the beam,

Table 4 Summary of dimensions and calculated values for each of the micro-beam tests

Test	L (μm)	b (μm)	h (μm)	I (μm^4)	P (μN)	E (GPa)
1	21.69	9.49	8.80	539	290	9.1
2	21.37	7.36	9.23	483	1016	30.4
3	24.12	7.25	9.47	514	353	14.2
4	23.18	7.95	9.94	651	478	13.5

yet the beam continued taking more load and exhibiting strain-hardening until a total failure.

This strain-hardening is typical of metals, including gold and tin, and cross-linked polymers when loaded in tension. These observations are unique to the progress in nanotechnology in material science fields, and they may only be observed in FIB–SEM micro/nanomechanical loading. Why at the nanometer scale these observations are unique? If we look at other natural nanocomposite materials such as bone, tooth, and shells, their micro- to macroscale mechanical behaviors are very similar to our KRS microbeams [30]. The ductile behavior observed in strain-softening and strain-hardening regimes is very convincing, when we evoke the granular porous material behavior and the pure elastomer polymer behavior as a nanocomposite at the micro-scales. Many earlier studies have also shown the intertwined nature of these components from micro- to macroscales [48, 54].

Finally, the brittle failures of T3 and T4 are not at all surprising to our community, since this is a typical behavior of a granular porous material failing in tension with a single fracture as in Fig. 18 frame (d). There is little to elaborate on, since they are typical force–displacement plots of a brittle granular material loaded to failure without any confining stress. In summary, Fig. 6 is a good representation of a source shale exhibiting tensile strength from Level 0 up to Level III, and into the macroscale of field applications.

6 Conclusions

This work had shed new light on the composite nature of kerogen-rich shale tensile failure. In hydraulic fracking, where Mode I fracture opening is behind the process of fracture propagation, i.e., rock failure subjected to tensile forces (tensile stresses), kerogen acts as a binding agent with a huge modulus of toughness. These experimental results showed that the composite nature of the organic-rich shale has tensile strength characteristics that are much more relevant and problematic to our field applications. An obvious question is why for the past decade or so in rock mechanics testing we did not pick up, on the tensile attributes of this KRS Woodford or any other source shale rock formation? The answer is simply that these tensile characteristics of polymers are easily masked in the ASTM and ISRM standard testing methods for macroscale geomechanics material

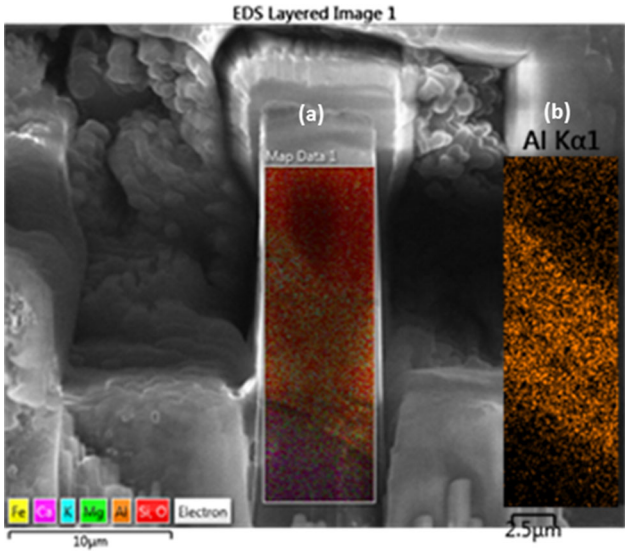


Fig. 20 **a** SEM images of the micro-pillar pre-loading (overlaid with the EDS map) and **b** EDS of the micro-pillar displayed and superimposed

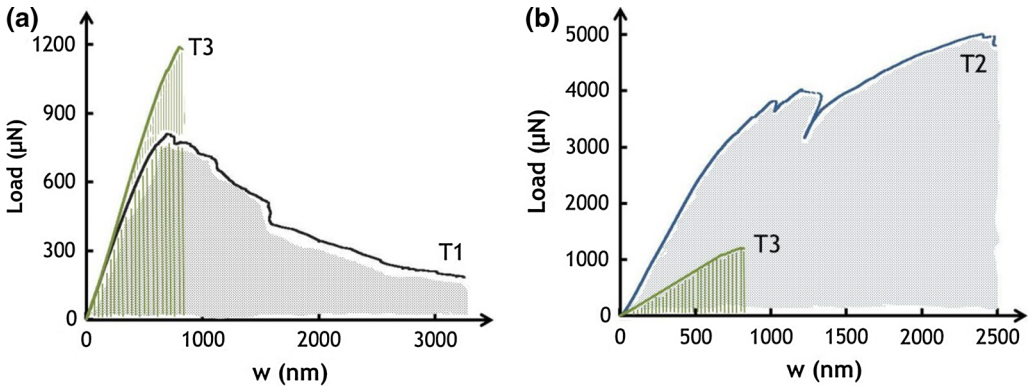


Fig. 19 Moduli of toughness for "granular shale" T3 when compared to the "kerogen elastomer cross-linked polymer" in T1 and T2

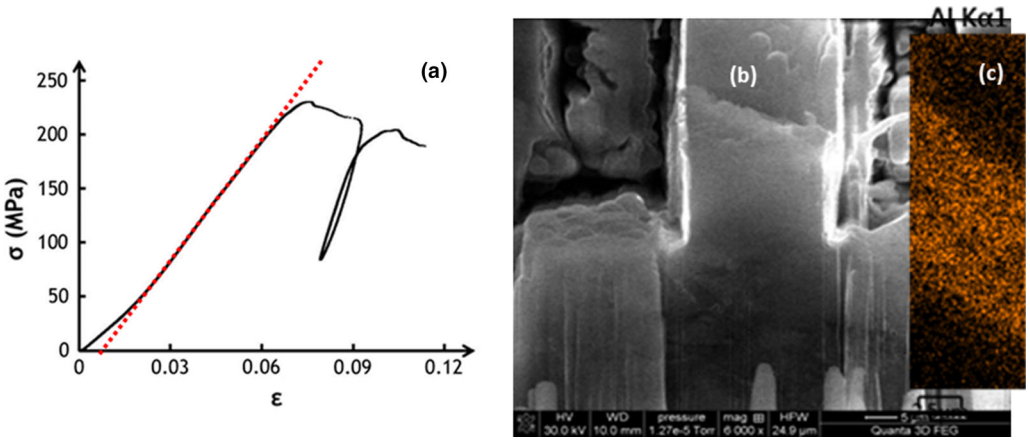


Fig. 21 **a** Plot of the stress versus strain, **b** SEM images of the micro-pillar after failure, only the *bottom left* is shown and **c** EDS map of the full micro-pillar superimposed showing the intact shear band plan pre-failure

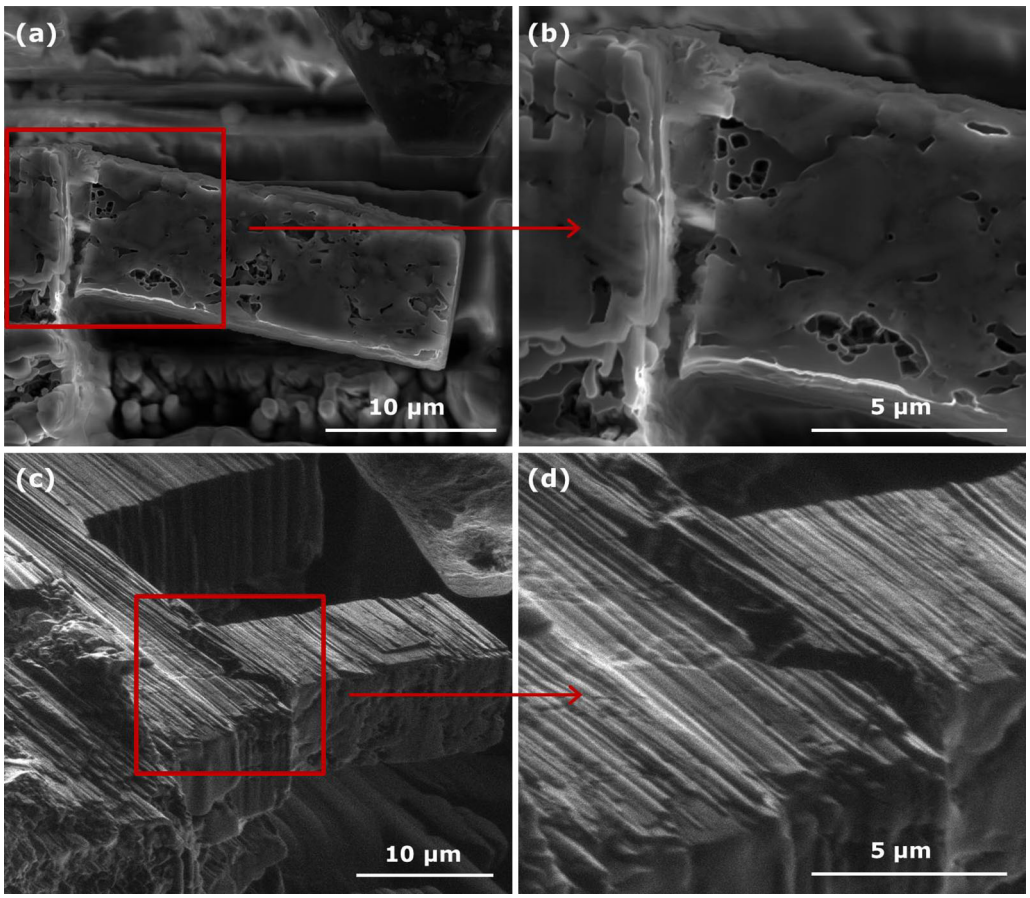


Fig. 22 SEM images of failed micro-beams, including (**a**) and (**c**) for Test 1, where the *red squares* indicate the sections that are zoomed inside the SEM in the subsequent frames (**b**) and (**d**), respectively. Notice in frame (**c**) the indenter tip held above the failed micro-beam, and yet it is still hanging at the support

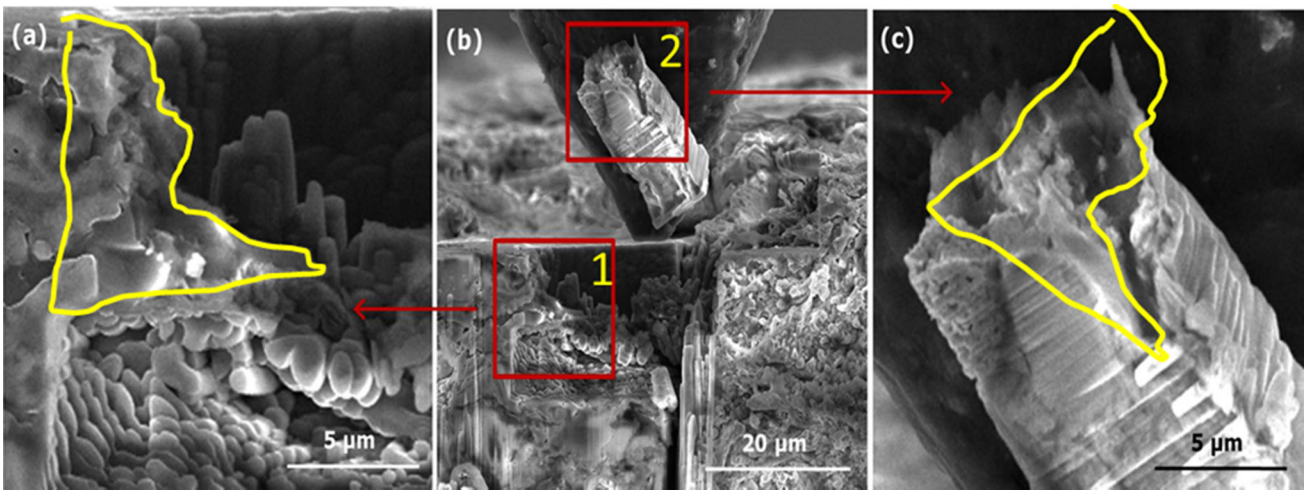


Fig. 23 SEM images of failed micro-beams Test 2, where the *red squares*, 1 and 2, in frame (**b**) indicate where the micro-beam broke from its support, the beam was stuck onto the indenter in this test. In frame (**a**) zoom on *red square* 1, to show the chunk of kerogen, contoured, that stayed behind as part of the beam support. In *red square* 2 is zoomed in frame (**c**) showing where that chunk of kerogen was part of the micro-beam originally and in frame (**b**) *red square* 2, we have it missing

characterization, such as the Brazilian test and other recommended tensile strength measurements for rocks. These tests were never designed to isolate or measure the tensile strength of polymers. This natural cross-linked polymer component, kerogen, with its tensile characteristics was not known previously to contribute to the tensile strength of any known shale loaded in tension. Now the organic-rich source shale formations are loaded under tensile forces, i.e., Mode I crack opening and crack propagation, and the ultimate tensile strength, UTS, of the organic components becomes of paramount importance to successfully engineer our field applications and redesign shale mechanical characterization for a wide range of modes of failures.

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Appendix

In Fig. 24, the micro-beam cantilever test, T1, is herein detailed for the load–displacement curve. The linear elastic early performance followed by the various slopes in the strain-softening regimes extended the micro-cantilever beam rupture to a very large displacement compared to the 600 nm for the early pure linear elastic deformation. From real-time in the SEM visualization, the dashed line

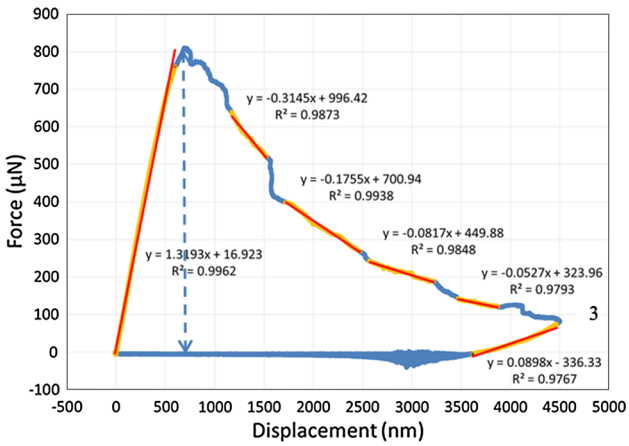


Fig. 24 Linear elastic load and rebound curves, in addition to the step-wise linear strain-softening behavior plotted in details

represents the first major fracture observed at the fixed support at a load of 809 µN. In other words, the kerogen after that point was supporting most of the load, thus preventing the beam from reaching its rupture strength. The rebound slope at the bottom after stage 3 shows a linear elastic rebound in the figure. This is the proof that the kerogen cross-linked elastomer nature did not reach its rupture strength, but rather that mass of kerogen extended the initial shale granular deformation and failure by almost 10 times to 4500 nm and still reserving an elastic rebound. Not to overload the paper, micro-beam in Test 2, T2, in strain-hardening (KRS micro-beam) could also be easily explained in a similar way in the loading–unloading and when brought eventually to failure.

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